

Samuel Karanja

✉ karsamanja@gmail.com

☎ (+254) 112292847

📍 Kenya

Professional Summary

Mathematics and AI specialist with strong foundations in machine learning, data science, and quantitative analysis, currently pursuing an MSc in Financial Engineering. Experienced in building real-time AI systems, automation workflows, and data-driven solutions with measurable business impact. Proven ability to evaluate models, analyze complex datasets, and translate business requirements into scalable technical solutions well-aligned with AI evaluation and quality assurance roles.

WORK EXPERIENCE

Techsari AI

AI Solutions Architect

01/01/2024 - Current

- Worked with stakeholders to assess current processes and transform business requirements into data-driven systems of scalable automation.
- Promised and executed the smart workflows and automation systems (scripts, bots, low-code tools) to minimize the manual effort, enhance the accuracy, and shorten the turnaround time.
- Corresponding data analysis and performance tracking to determine the effectiveness of automation and provide quantifiable efficiency and performance improvement.
- Constant optimization of automation pipelines through KPI checkups and system optimization on actual performance findings.

EcoOffgrid

Business Continuity Analyst

03/02/2025 - 29/09/2025

- Served as an automation specialist, designing data-driven workflows that reduced manual data entry by **85%** and significantly improved operational efficiency
- Analyzed business processes and identified opportunities to streamline workflows using automation and scalable system design
- Developed and implemented automation solutions that accelerated the sales cycle and enhanced overall process reliability
- Evaluated system performance and workflow efficiency, contributing to continuous improvement and measurable productivity gains
- Advised on technology stack optimization and automation strategies to support scalable growth in the renewable energy sector

EyeKnow AI

Machine Learning Engineer

01/05/2022 - 19/09/2022

- Developed and deployed a real-time computer vision system for violence detection, achieving **90% accuracy** in live environments
- Applied deep learning techniques to process and analyze video data, enabling reliable detection under varying real-world conditions
- Designed and implemented an automated monitoring and alert system that delivered real-time notifications to security personnel, reducing response time
- Evaluated model performance and continuously improved detection accuracy through testing, tuning, and data-driven optimization
- Collaborated with cross-functional teams to integrate the solution into operational environments, ensuring scalability and reliability

Renewsenses

Machine Learning Engineer

01/06/2021 - 30/11/2021

- Engineered a real-time object recognition model to detect arriving buses, accurately accommodating diverse conditions and international vehicle designs.
- Implemented an OCR solution that precisely reads bus numbers and routes, empowering users to identify the correct transport independently.
- Designed an object detection model to locate vacant seats with high reliability, enhancing passenger accessibility and safety.

Weedbot

Linked Data Engineer

01/12/2020 - 29/03/2021

- Developed real-time object detection and OCR systems for public transport using deep learning
- Improved accessibility by enabling accurate bus and route identification for users
- Evaluated and optimized model performance for reliability in real-world conditions

EDUCATION

WorldQuant University

MASTER OF SCIENCE IN FINANCIAL ENGINEERING - 8

07/10/2025 - Current

- Developed a solid background in quantitative finance, including derivatives pricing, portfolio management, risk management, and statistical modeling.
- The use of mathematical and statistical techniques for practical financial challenges emphasizes the use of data to analyze and make decisions.
- Gained hands-on experience with Python and quantitative approaches to time series analysis and algorithmic strategies.
- Acquired expertise in statistical analysis of complicated datasets to model trends, assess risk, and create actionable information that applies to economic and policy environments.

Kenya Methodist University

BACHELOR OF SCIENCE IN MATHEMATICS AND COMPUTER SCIENCE - 6

07/05/2018 - 19/10/2024

- Developed a good background in mathematics, such as calculus, linear algebra, probability and statistics, and the fundamentals of other computer science fields, such as algorithms, data structures, and databases.
- Gained advanced experience in Python, Java and JavaScript programming languages, having acquired and applied problem-solving and effective system design.
- Use quantitative and analytical tools to address complex problems, aid in the analysis of data and informative research.
- Acquired practical knowledge in software development and data manipulation, dealing with structured data to aid in the decision-making process in economics and policy-based issues.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn, FastAPI), Git, Docker, Linux, REST APIs

Data Science, AI & Machine Learning: Machine Learning, Deep Learning, Recommender Systems, Statistical Software Development, Data Science

Data & BI: SQL, Data pipelines, EDA, KPI design, business intelligence, and reporting

Systems & soft skills: End-to-end AI system design, automation, clear communication with technical and non-technical stakeholders

CERTIFICATE

DeepLearning.AI TensorFlow Developer

Natural Language Processing in TensorFlow

Neural Networks and Deep Learning

Deep Learning with TensorFlow

Deep Learning using TensorFlow

Google Data Analytics

Exploratory Data Analysis with Python and Pandas

AI Product Management Specialization

Managing Machine Learning Projects

Project Management: Communication, Stakeholders & Leadership